

Maths

Probability

1. Probabilistic graphical models can only do 2 things:
 1. Inference: calculating a probability of some data modelled as RV using the model $\mathcal{M}$:
 1. evaluation $P(obs|state)$,
 2. decoding $P(state|obs)$
 2. Learning model parameters: refers to finding the graphical model $\mathcal{M}$, given some data $\mathcal{D}$.
 1. MAP (maximum a posteriori)
 2. Maximum likelihood
 3. expectation maximization
 4. forward-backward algorithm
2. Conditional probability can be computed using:
 1. Bayes rule
 2. Enumeration
3. θ is model parameter.
4. For example RV x is binomial, x is the number of times a coin landed heads in N trials.

1.
$$P(x = X|\theta = p) = \binom{N}{x} p^x (1-p)^{N-x} \quad (\text{likelihood})$$

5. if RV x is single variable Gaussian, for modeling a real value, e.g. temperature sensor reading.

1.
$$P(x = X|\theta = (\mu, \sigma^2)) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} \quad (\text{likelihood})$$

6. Note: Notice that in both cases the RV x ended up in the exponent.

7.
$$p(\theta|x = D) = \frac{p(x|\theta)p(\theta)}{p(x)}$$

$p(x|\theta)$ is the likelihood, $p(\theta)$ is the prior distribution, $p(\theta|x = D)$ is the posterior, and $p(x)$ is the normalization term.

References:

1. Harrison H. Barrett, Kyle J. Myers. Foundations of Image Science.
2. [Neural Message Passing](#)